Selected Numerical Results for the Hybrid LSMC–PDE Method

BGK-style calibrated gDMR model with 12 Bermudan exercise dates

27 March 2026

1. Experimental Setting

The experiments in this note use the BGK-style calibrated gDMR specification

$$S_0 = 100, \quad v_0 = 0.114, \quad v_0' = 0.110, \quad r = 0, \quad T = 1,$$

with

$$\kappa_1 = 5.5, \quad \kappa_2 = 0.1, \quad \theta = 0.078, \quad \xi_1 = 2.689, \quad \xi_2 = 0.502,$$

and

$$\delta_1 = \delta_2 = 0.94, \quad \rho_{12} = -0.982, \quad \rho_{13} = -0.727, \quad \rho_{23} = 0.59.$$

The Bermudan contract is sampled at 12 exercise dates. Three strike levels are considered:

$$K \in \{90, 100, 110\},$$

corresponding to OTM, ATM, and ITM puts.

The benchmark method is standard LSMC. The hybrid method couples LSMC with a PDE correction on a one-dimensional asset grid. In the tuned hybrid runs reported below, the common numerical setting is

$$\text{paths} = 60{,}000, \quad \text{asset points} = 301, \quad S_{\min}/S_0 = 0.30, \quad S_{\max}/S_0 = 3.50,$$

with volatility truncation quantile 0.999.

For any direct price estimate $\widehat{V}$ and reference value V^{ref}, the direct relative error is defined by

$$\varepsilon_{\mathrm{dir}} = \frac{|\widehat{V} - V^{\mathrm{ref}}|}{|V^{\mathrm{ref}}|}.$$

2. Tuned Comparison Across Moneyness

The first experiment fixes 12 exercise dates and compares the tuned hybrid method with the benchmark LSMC under the original BGK-style production setting with 600 Euler steps. The benchmark uses 10^6 paths, while the tuned hybrid uses 60,000 paths.

Table 1: Direct-price comparison under the tuned hybrid setting.

Scenario	K	Benchmark direct	Hybrid direct	Hybrid dir. err.
ATM	100	11.371204	11.359836	0.100%
ITM put	110	16.637388	16.680641	0.260%
OTM put	90	7.426156	7.414778	0.153%

The tuned hybrid method achieves direct relative errors below 0.3% in all three moneyness regimes. In particular, the ATM and OTM cases are reduced to 0.100% and 0.153%, respectively, while the ITM case remains at 0.260%.

3. Path-Count Study at 48 Euler Steps

To study the direct estimator under a coarser time discretization, a path-count sweep was run with Euler steps fixed at 48. For this comparison, the reference prices were strengthened by using benchmark-only runs with 1200 Euler steps and 1,200,000 benchmark paths:

$$V_{\mathrm{ATM}}^{\mathrm{ref}} = 11.356823, \qquad V_{\mathrm{OTM}}^{\mathrm{ref}} = 7.402999.$$

The results are especially favorable to the hybrid method. For both ATM and OTM, the hybrid direct error remains below the LSMC benchmark error at every tested path count from 250 to 60,000.

Table 2: Representative direct relative errors at 48 Euler steps, rebased to the 1200-step benchmark reference.

Scenario	Paths	Benchmark dir. err.	Hybrid dir. err.
ATM	1,000	6.053%	1.620%
ATM	10,000	1.806%	0.960%
ATM	20,000	1.336%	0.052%
ATM	60,000	1.232%	0.302%
OTM put	1,000	7.840%	2.905%
OTM put	10,000	2.877%	0.869%
OTM put	20,000	2.173%	0.248%
OTM put	60,000	1.769%	0.526%

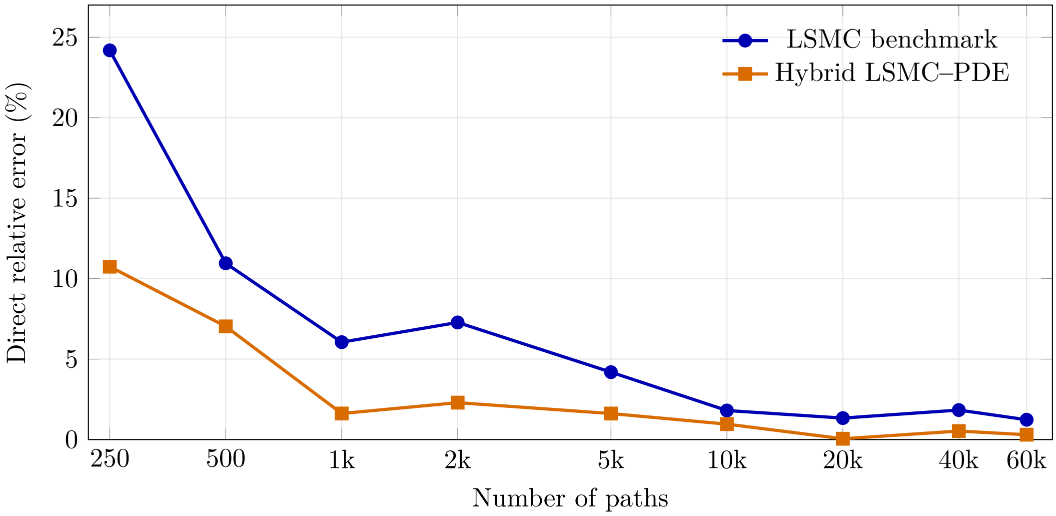

Figure 1: ATM direct relative error at 48 Euler steps, rebased to the 1200-step benchmark reference.

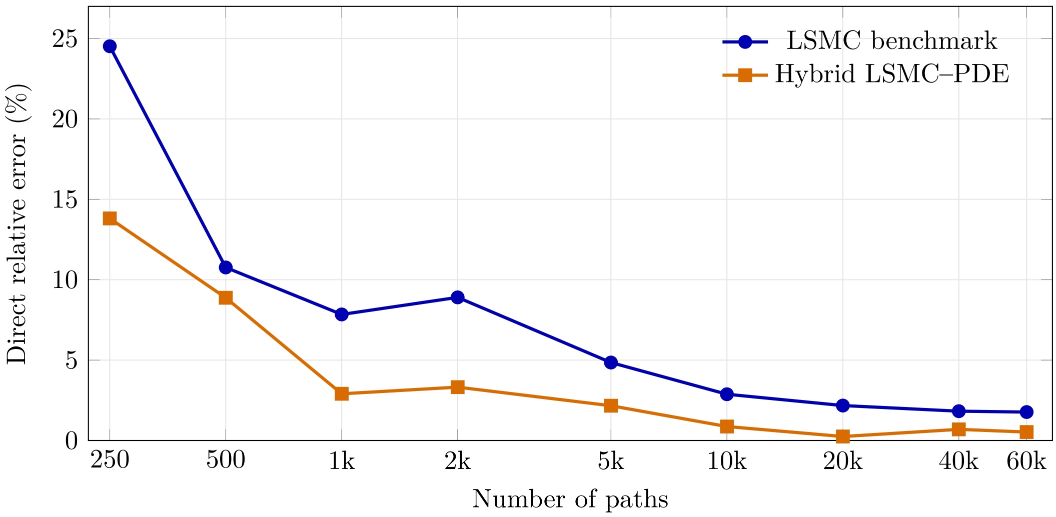

Figure 2: OTM direct relative error at 48 Euler steps, rebased to the 1200-step benchmark reference.

4. Additional Check at 60 Euler Steps

An additional saved comparison is available at 60 Euler steps. Rebased to the same 1200-step benchmark references, the hybrid method still produces smaller direct errors for both ATM and OTM:

Table 3: Direct relative error at 60 Euler steps under the stronger benchmark reference.

Scenario	Benchmark direct	Hybrid direct	Benchmark dir. err.	Hybrid dir. err.
ATM	11.381616	11.350391	0.218%	0.057%
OTM put	7.435732	7.410080	0.442%	0.096%

Although the hybrid advantage is strongest at 48 Euler steps, the saved 60-step check indicates that the improvement in direct pricing accuracy persists when the time grid is moderately refined.

5. Conclusion

The experiments reported here support the following conclusions.

- With 12 exercise dates and the tuned numerical setting, the hybrid method delivers accurate prices across ATM, ITM, and OTM cases, with direct relative errors below 0.3%.
- At a coarse Euler grid of 48 steps, the hybrid direct estimator is uniformly more accurate than the benchmark LSMC estimator over the full tested path range in both ATM and OTM cases, even when the reference is strengthened to a 1200-step, 1.2 million-path benchmark.
- The saved 60-step comparison remains favorable to the hybrid method, suggesting that the improvement is not confined to the most extreme coarse-grid regime.

Taken together, these results indicate that the hybrid LSMC–PDE method is particularly effective when high-quality direct prices are required under a relatively coarse time discretization, while still maintaining good accuracy across different moneyness regimes.